

SDLC Overview

① Requirements

- Purpose: Gather and analyze what the client needs
- Importance: Defines what is to be built; Sets the foundation for the project.
- Output: Requirement Specification Document
- Keywords: Stakeholder interviews, use cases, feasibility study.

② Design

- Purpose: Plan how the software will be built
- Importance: Translates requirements into a blueprint.
- Output: System Architecture, UI/UX designs, Database models.
- Keywords: UML diagrams, flowcharts, design patterns.

③ Implementation (coding)

- Purpose: Developers write code based on design specs.
- Importance: The actual creation of the software product.
- Output: Source code, version control updates.
- Keywords: IDEs, Git, programming languages

④ Testing

- Purpose: Ensure the software works as intended
- Importance: Identifies bugs and verifies quality.
- Output: Test cases, bug reports, QA approval
- Keywords: Unit testing, integration testing, system testing.

⑤ Deployment

- Purpose: Release the product to users
- Importance: Delivers the working software to production.
- Output: Live product, user documentation
- Keywords: CI/CD, release management, rollout plans.

Interconnection:

→ Each phase flows into the next; however, feedback loops exist.

→ Example: Testing may lead back to implementation.

→ Agile & DevOps emphasize continuous integration between phases.